

Getting Started Guide

Configure PingOne Advanced Identity Cloud (Ping AIC) / ForgeRock Identity Cloud with LexisNexis One-Time Passcode (OTP) Nodes

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SCOPE

LexisNexis Risk Solutions (LNRS) Dynamic Decision Platform (DDP) hosts the Authentication Hub's one-time passcode (OTP) product. LexisNexis OTP is an out-of-band authentication method that provides business and government organizations the ability to have stronger authentication during a high risk, high value transaction with a customer. This document contains the detailed steps and supporting information required to install and configure PingOne Advanced Identity Cloud (Ping AIC), formerly ForgeRock Identity Cloud with LexisNexis OTP Nodes. This guide is intended to install a simple configuration to support testing.

The following are architecture assumptions and limitations:

- Ping AIC / ForgeRock Identity Cloud tenant is configured
- Ping AIC / ForgeRock Identity Cloud has default configuration for the Identity Store where the Alpha realm will be utilized for the test user
- A test account has been configured via Identities that includes first name, last name, email address, mobile phone, home phone, and login username

Document Organization

This document is divided into four sections as follows:

- **Scope.** Defines the purpose of this document.
- **LexisNexis OTP Nodes.** Provides an overview of the nodes available for authentication journeys.
- **LexisNexis Dynamic Decision Platform Portal.** Provides detailed information regarding the configuration of LexisNexis Dynamic Decision Platform (DDP) to include configuration of a simple policy and how to access configuration parameters required for the authentication journey.
- **Authentication Journey Configuration.** Provides detailed steps to configure a ForgeRock journey with LexisNexis OTP Nodes to perform Multi-Factor Authentication (MFA).

LEXISNEXIS ONE-TIME PASSCODE NODES

LexisNexis One Time Password is an out-of-band authentication method that provides business and government organizations the ability to have stronger authentication during a high risk, high value transaction with a customer. It offers a time-sensitive, unique random passcode via SMS text, email or phone and is ideal for companies that are interested in providing a multi-factor authentication solution for their customers. No hardware (electronic fob, etc.) other than the user's existing phone or personal computer is required.

To include a LexisNexis OTP node in an authentication journey, enter LexisNexis into the Filter nodes to get a listing of available capabilities. For OTP, the following nodes are available:

- LexisNexis OTP Selector
- LexisNexis OTP Send
- LexisNexis OTP Collector
- LexisNexis OTP Decision

LexisNexis OTP Selector

This node will display a list of selections to the user for available methods to send an OTP code. LexisNexis OTP Nodes provide for email/OTP, SMS/OTP and voice/OTP. Upon user selection, the node will place the selected value into shared state variable `otp.type_selected`. By default, this is the shared state variable that the LexisNexis OTP Send node accepts to determine which type of OTP to send to the end user. If only a single OTP type is configured within the LexisNexis OTP Selector, the node just sends the `otp.type_selected` without any user interaction.

This node is optional within an authentication tree/journey. The node is intended as a helper for an orchestration to retrieve the end user desired OTP type. The LexisNexis OTP Send node just requires the field `otp.type_selected` which can be placed into shared state by any other node in the authentication tree/journey.

Input

The LexisNexis OTP Selector node does not have any required/optional input attributes.

Configuration

The LexisNexis OTP Selector node has the following configuration parameters:

- **Send Email** - When selected, this will display an Email Button on the selector interface.
- **Send SMS** - When selected, this will display an SMS Button on the selector interface.
- **Send Voice** - When selected, this will display an Voice Button on the selector interface.

Output

The LexisNexis OTP Selector node has the following outputs placed into shared state:

- **OTP Type Selected** - The user selected OTP type will be contained in the state variable `otp.type_selected`. This String value is further processed by the LexisNexis OTP Send node.
- **Failure Reason** - If an error outcome is generated, the state variable `otp.failure_reason` will contain text associated with the error condition.

Callbacks

The LexisNexis OTP Selector node has the following callbacks:

- When more than one OTP Type is configured, the following callbacks for the user interface are returned:
 - **ConfirmationCallback** - Contains a button for each active configured OTP Type. The label in the button is based on the configured String value.

Outcomes

The LexisNexis OTP Selector node has the following outcomes:

- **Success** - This outcome is triggered when a valid OTP Type is selected by the user.
- **Error** - This outcome is triggered when there is a fundamental integration error. First attempt to fix the integration error by looking at debug log files for the node to determine if the integration error is due to configuration. If the configuration looks accurate, then open a support case with LexisNexis Risk Solutions.

LexisNexis OTP Send

This node will send an API request to the LexisNexis Dynamic Decision Platform (DDP) authentication hub that sends an OTP code to the end user. There is no interface displayed to the user by this node. The OTP Type selected attribute in shared state variable `otp.type_selected` defines the delivery method, along with the attribute configuration to locate the destination (e.g. user email address or phone number). The Attribute Source defines where the attribute mapping information will be fetched at runtime, mainly either User Directory or Shared State. When the user directory is configured as the source for attributes, the node will assume that the username is contained in shared state from a previous node in the authentication tree/journey and use that username to query the user directory for the user's destination. When the shared state is configured as the source for attributes, the node will inspect shared state for the user's destination.

Input

The LexisNexis OTP Send node retrieves the following from the journey shared state based upon the configuration of the Attribute Source parameter.

- **OTP Type Selected** - Required in shared state variable `otp.type_selected` to define the OTP delivery method. The LexisNexis OTP Selector node can be included in a authentication tree/journey to place this state variable into shared state, or it can be written by another node.
- **AM Username (username)** - Required in shared state, or `objectAttributes`, to resolve the desired AM identity when the Attribute Source = User Directory. The AM identity user record is queried for the OTP destination (e.g. user email address or phone number).
- **objectAttributes** - Required when the Attribute Source = Shared State. Shared state is queried for the OTP destination (e.g. user email address or phone number). The attribute to query is based on the configuration parameter **Email Attribute**, **SMS Attribute**, or **Voice Attribute**.

Configuration

The LexisNexis OTP Send node has the following configuration parameters:

- **Org ID** - Org ID is the unique id associated your organization on the Dynamic Decision Platform (DDP).
- **API Key** - This is the unique API key generated via DDP Portal associated to the Org ID.
- **Base URL** - Defines the domain URL for the DDP/TMX region where API Requests are to be sent. The default value is the global region.
- **Policy** - The DDP Portal policy to be used to integrate the DDP Authentication Hub with OTP.
- **OTP Length** - Length of the OTP. Valid values are between 6 and 10 characters.
- **OTP Expire** - Expiration time in minutes for the OTP. Valid values are between 1 and 60 minutes.
- **Email Title** - When Email/OTP is triggered, this will be the title of the email sent to the user.
- **Email Message** -When Email/OTP is triggered, this will be the message body of the email sent to the user.
- **SMS Message** - When SMS/OTP is triggered, this will be the message body of the SMS text message sent to the user. The SMS message has a maximum length of 160 characters.
- **Attribute Source** - Defines where the attributes for sending the OTP code is fetched at runtime. This is a dropdown list that contains the options User Directory and Shared State. User Directory will look for attribute in the Identity Store, and Shared State looks in the shared memory.
- **Email Attribute** - When Email/OTP is defined by the OTP Type, the attribute defined in this configuration parameter will be fetched by the name of the attribute provided based on the Attribute Source defined.
- **SMS Attribute** - When SMS/OTP is defined by the OTP Type, the attribute defined in this configuration parameter will be fetched by the name of the attribute provided based on the Attribute Source defined.
- **Voice Attribute** - When Voice/OTP is defined by the OTP Type, the attribute defined in this configuration parameter will be fetched by the name of the attribute provided based on the Attribute Source defined.

Output

The LexisNexis OTP Send node has the following outputs placed into shared state:

- **OTP API Request ID** - The Request ID (e.g. attribute `request_id`) associated to the DDP Authentication Hub API Response for OTP will be placed into state variable `otp.request_id`. This String value is used by the LexisNexis OTP Decision node to invoke an API Request with OTP passcode entered by the user for validation.
- **OTP Sent To** - Contains the user's email address or phone number based on OTP delivery method. The information is stored in state variable `otp.sent_to` and displayed by the LexisNexis OTP Collector node as part of the user interface.
- **OTP Policy** - Contains the DDP Policy that invoked the OTP Auth Hub. The state variable `otp.policy` places the policy name into shared state for the LexisNexis OTP Decision node to consume, which is a requirement for the same policy to be used.
- **OTP URL** - Contains the DDP URL that was configured in this node. The state variable `otp.url` is used by the LexisNexis OTP Decision node to send the API request for validation.
- **Failure Reason** - If an error outcome is generated, the state variable `otp.failure_reason` will contain text associated with the error condition.

Callbacks

The LexisNexis OTP Send node does not have any callbacks as there is no user interface displayed.

Outcomes

The LexisNexis OTP Send Node has the following outcomes:

- **Success** - This outcome is triggered when the OTP code is successfully generated for the user. It is worth mention that generation does not guarantee delivery to the users device. Thus, the LexisNexis OTP Collector Node allows for retry in the event the user never receives the OTP code.
- **API Error** - This outcome is triggered when there is an issue with the API Request such as a network timeout or the service is unavailable.
- **OTP Fail** - This outcome is triggered when the API Request is rejected by the LexisNexis DDP Authentication Hub service. Within the debug logging for the node, the actual error codes will be written for offline analysis and triage of the issue.
- **Error** - This outcome is triggered when there is a fundamental integration error. First attempt to fix the integration error by looking at debug log files for the node to determine if the integration error is due to configuration. If the configuration looks accurate, then open a support case with LexisNexis Risk Solutions.

LexisNexis OTP Collector

This node collects the OTP code sent to the user. The interface enables the user to submit an OTP code for decision and validation, as well as the user can request the OTP code to be resent.

Input

The LexisNexis OTP Collector node retrieves the following from journey shared state:

- **OTP Type Selected** - Required in shared state variable `otp.type_selected` to define the OTP delivery method. This nodes uses the value to customize the user interface message prompting for OTP Passcode entry.
- **OTP Sent To** - Contains the user's email address or phone number based on OTP delivery method. The information is stored in state variable `otp.sent_to` and displayed by the LexisNexis OTP Collector node as part of the user interface.
- **OTP Error** - State variable `otp.error` contains a code to define whether or not an error message should be displayed to the user. Values include False, Blank and True.

Configuration

The LexisNexis OTP Collector Node has the following configuration parameters:

- **Message Body** - This is the message displayed to the user on the collector interface. The variable `${otpDestination}` will contain either an email address or phone number depending on the method selected by the user via the Lexis OTP Sender.
- **Help Text** - This is the help text displayed in the OTP text entry box for the user.
- **OTP Error Message** - Message displayed if user enters an incorrect OTP code
- **OTP Blank Message** - Message displayed if user attempts to submit a blank OTP code

Output

The LexisNexis OTP Collector node has the following outputs placed into shared state:

- **OTP Passcode** - Contains the user submitted OTP passcode for validation. The associated String is placed into state variable `otp.passcode_entered`. This value is further processed by the LexisNexis OTP Decision node.
- **Failure Reason** - If an error outcome is generated, the state variable `otp.failure_reason` will contain text associated with the error condition.

Callbacks

The LexisNexis OTP Collector node has the following callbacks:

- **TextOutputCallback** - When an error occurs and needs to be displayed to the user, this first **TextOutputCallback** container will contain the error message to display to the user. The container will be marked as `TextOutputCallback.ERROR`. The following state variable `otp.error` values indicate that the object should be created:
 - **Blank** - Fills the text based on the **OTP Blank Message**
 - **True** - Fills the text based on the **OTP Error Message**
- **TextOutputCallback** - Contains the configured **Message Body**
- **NameCallback** - Container to collect the user entered OTP Passcode. The configured **Help Text** is displayed in the text entry box.
- **ConfirmationCallback** - Contains buttons for the user interface, mainly Submit and Retry buttons.

Outcomes

The LexisNexis OTP Collector Node has the following outcomes:

- **Submit** - This outcome is triggered when the user selects the “Submit” button. When selected, this should then link to the LexisNexis OTP Decision node to validate the OTP Code being submitted. If the LexisNexis OTP Decision node detects a blank or invalid OTP code, then this node will be linked from the decision node to re-prompt the user for OTP code entry, as well as display the appropriate error message. The outcomes review and challenge from the LexisNexis OTP Decision are to link back to the LexisNexis OTP Collector.
- **Retry** - This outcome is triggered when the user wants to get a new OTP code by selecting the “Retry” button on the interface. This outcome should route back to the beginning of the OTP user workflow.
- **Error** - This outcome is triggered when there is a fundamental integration error. First attempt to fix the integration error by looking at debug log files for the node to determine if the integration error is due to configuration. If the configuration looks accurate, then open a support case with LexisNexis Risk Solutions.

LexisNexis OTP Decision

This node verifies OTP codes entered by the user. In a typical authentication tree, the LexisNexis OTP Collector will precede this node. The collector places the user entered and submitted OTP in shared state. Additionally, the LexisNexis OTP Sender will precede this node and the LexisNexis Collector Node that places the characteristics of the type of OTP into shared state.

Input

The LexisNexis OTP Decision node retrieves the following from the journey shared state:

- **OTP Passcode** - Contains the user submitted OTP passcode for validation. The associated String is placed into state variable `otp.passcode_entered`.
- **OTP API Request ID** - The Request ID (e.g. attribute `request_id`) associated to the DDP Authentication Hub API Response for OTP is fetched from state variable `otp.request_id`. This String value is used by the LexisNexis OTP Decision node to invoke an API Request with OTP passcode entered by the user for validation.
- **OTP Policy** - Contains the DDP Policy that invoked the OTP Auth Hub. The state variable `otp.policy` is used to invoke the API Request to DDP Auth Hub.
- **OTP URL** - State variable `otp.url` is used by the LexisNexis OTP Decision node to send the API request for validation.

Configuration

The LexisNexis OTP Decision Node has the following configuration parameters:

- **Org ID** - Org ID is the unique id associated your organization on the Dynamic Decision Platform (DDP).
- **API Key** - This is the unique API key generated via DDP Portal associated to the Org ID.

Output

The LexisNexis OTP Decision node has the following outputs placed into shared state:

- **Failure Reason** - If an error outcome is generated, the state variable `otp.failure_reason` will contain text associated with the error condition.

The LexisNexis OTP Decision node cleans up and removes data from shared state upon an outcome of pass, reject or error:

- **OTP Request ID** - The Request ID associated to the DDP Authentication Hub API Response from the IIDQA Get Quiz node contained state variable `otp.request_id`.
- **OTP Passcode** - Removes state variable `otp.passcode_entered`.
- **OTP Policy** - Removes state variable `otp.policy`.
- **OTP URL** - Removes state variable `otp.url`.
- **OTP Error** - Removes state variable `otp.error`.

Callbacks

The LexisNexis OTP Decision node does not have any callbacks as there is no user interface displayed.

Outcomes

The LexisNexis OTP Decision Node has the following outcomes:

- **Pass** - This outcome is triggered when the OTP code submitted by the user is validated so that MFA via OTP has passed.
- **Challenge** - This outcome is triggered when the OTP code fails to validate, and the number of retries has not been violated.
- **Review** - This outcome is triggered when the OTP code fails to validate due to blank OTP code entered.
- **Reject** - This outcome is triggered when OTP code validation is failed.
- **Error** - This outcome is triggered when there is a fundamental integration error. First attempt to fix the integration error by looking at debug log files for the node to determine if the integration error is due to configuration. If the configuration looks accurate, then open a support case with LexisNexis Risk Solutions.

DYNAMIC DECISION PLATFORM PORTAL CONFIGURATION

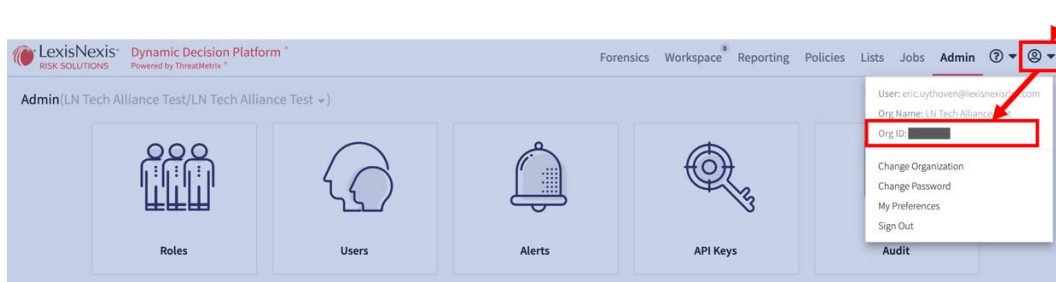
This section defines the high-level LexisNexis Dynamic Decision Platform (DDP) Portal configuration items that will be needed for the overall configuration. There are two main categories of configuration, mainly,

- Organization ID and API Key for the REST API interfaces. This information is needed by the LexisNexis OTP Nodes and will be entered as part of configuration.
- LexisNexis DDP Portal Policy. The configured policy within DDP provides the configuration for the Authentication Hub to access OTP services. For the purposes of the getting started guide and to have a simple test configuration for different outcomes, the “OTP” policy will be configured to directly integrate the Authentication Hub without any further policy rules.

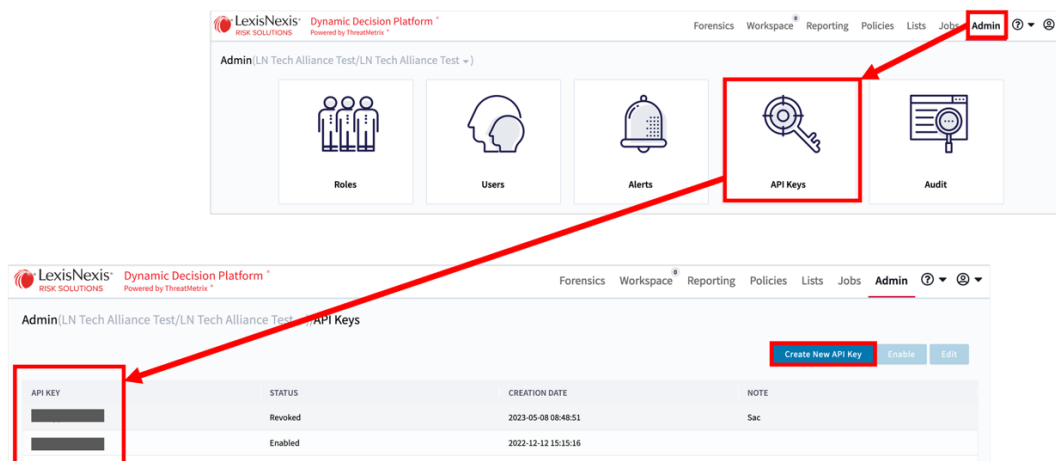
Retrieve OrgID and API Key

To retrieve the DDP Portal values for Organization ID and API Key, perform the following steps.

1. Access **DDP Portal** over the internet by logging into your administrative account with credentials provided by LexisNexis Risk Solutions.
2. From the **DDP Portal** home page, select the user information dropdown that will display username, OrgName and OrgID. This will be the OrgID to enter into the configuration of the LexisNexis OTP Nodes.



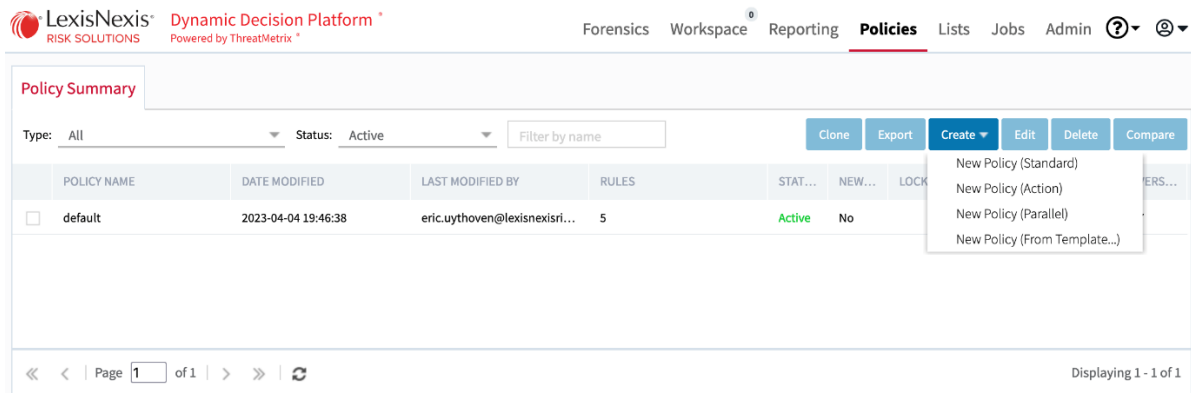
3. Within the **DDP Portal** home page, select **Admin** followed by selecting the **API Keys** tile. Retrieve the value for API Key. In the event no API Key is listed, select the **Create New API Key** button to generate a new key. This will be the API Key to enter into the configuration of the LexisNexis OTP Nodes. The API Key is to be protected. Do not email or keep this value in cleartext on any computer system.



Dynamic Decision Platform Portal Policy

For the purposes of the getting started guide and to have a simple test configuration for different outcomes, the “OTP” policy will be configured to directly integrate the Authentication Hub without any further policy rules. Perform the following steps.

1. Access **DDP Portal** over the internet by logging into your administrative account with credentials provided by LexisNexis.
2. From the **DDP Portal** home page, select **Policies** from the menu bar. This will provide a listing of available policies that can be configured with the LexisNexis OTP Nodes. The first step is to select the **Create** dropdown menu followed by **New Policy (Standard)**.



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Forensics Workspace Reporting **Policies** Lists Jobs Admin ? @

Policy Summary

Type: All Status: Active Filter by name

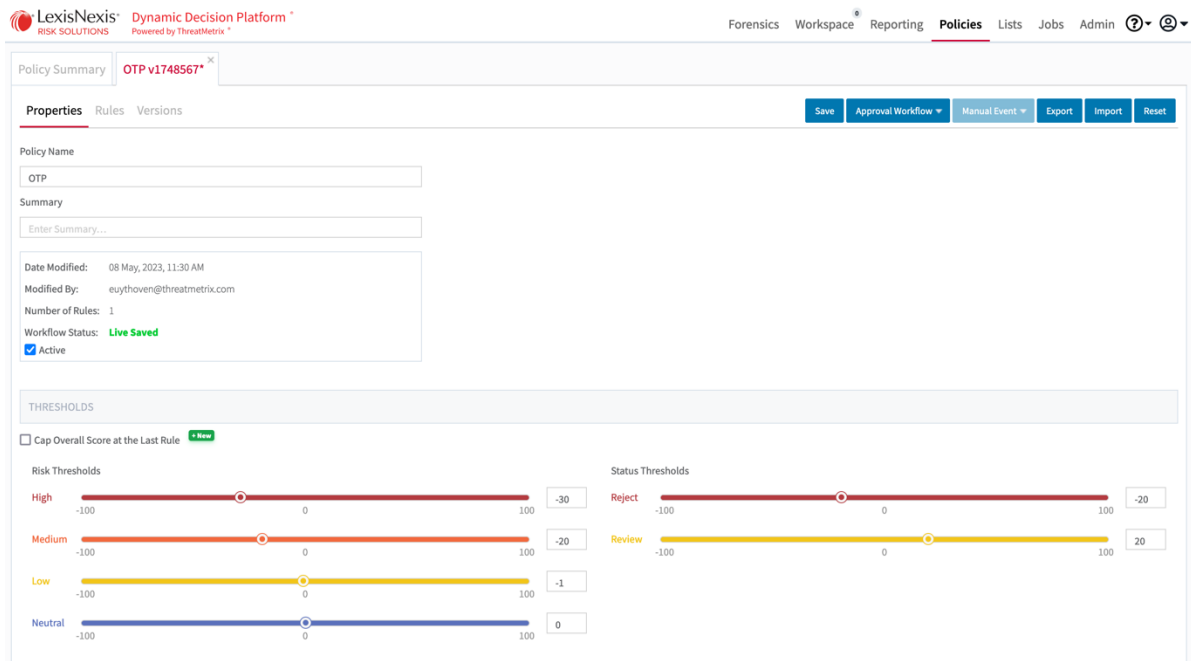
Clone Export Create Edit Delete Compare

	POLICY NAME	DATE MODIFIED	LAST MODIFIED BY	RULES	STAT...	NEW...	LOCK	ERS...
☐	default	2023-04-04 19:46:38	eric.uythoven@lexisnexisr...	5	Active	No		

« < | Page 1 of 1 | > »

Displaying 1 - 1 of 1

3. On the Policy Summary, the **Properties** interface tab will be displayed. Enter Policy Name = OTP, select the Active button, and update the Status Thresholds for Reject = -20 and Review = 20.



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Forensics Workspace Reporting **Policies** Lists Jobs Admin ? @

Policy Summary OTP v1748567

Properties Rules Versions

Save Approval Workflow Manual Event Export Import Reset

Policy Name: OTP

Summary: Enter Summary...

Date Modified: 08 May, 2023, 11:30 AM
Modified By: euythoven@threatmetrix.com
Number of Rules: 1
Workflow Status: Live Saved
☒ Active

THRESHOLDS

☐ Cap Overall Score at the Last Rule

Risk Thresholds

High: -100 0 100 -30

Medium: -100 0 100 -20

Low: -100 0 100 -1

Neutral: -100 0 100 0

Status Thresholds

Reject: -100 0 100 -20

Review: -100 0 100 20

- To create the policy rules, select the **Rules** interface tab. The OTP policy will be a single **Authentication** rule to integrate the Authentication Hub as follows.

Copy Selected
Paste
Delete
☐ Show Details

Enter search text...

NAME (REASON CODE)	RULE TYPE	DESCRIPTION	RISK WE...	RE...
OTP Auth	Authentication	Authentication Connector	0	Yes

- The **Authentication Rule Editor** can follow the template shown here. Within this interface, the **Product Configuration** is the Authentication Hub configuration that is established via LexisNexis Professional Services. The services configure the LexisNexis OTP service as an interface associated to the customer account.

Authentication Rule Editor
X

Name *
OTP Auth

Summary

☒ Generate Reason
☐ Generate Summary Reason

ESB Mode
☒ Published
☐ Test

Select Authentication Strategy
Product Configuration *
TechAlliance_OTP

☒ Terminate Policy After This Rule

Actions:
Action Templates can be specified here and will be triggered at the end of the full authentication flow based on the final outcome.
Only configured Action Templates for your org will appear here.

Authentication Success
Challenge Pass

Authentication Failure
Default Reject

Timeout
Default Reject

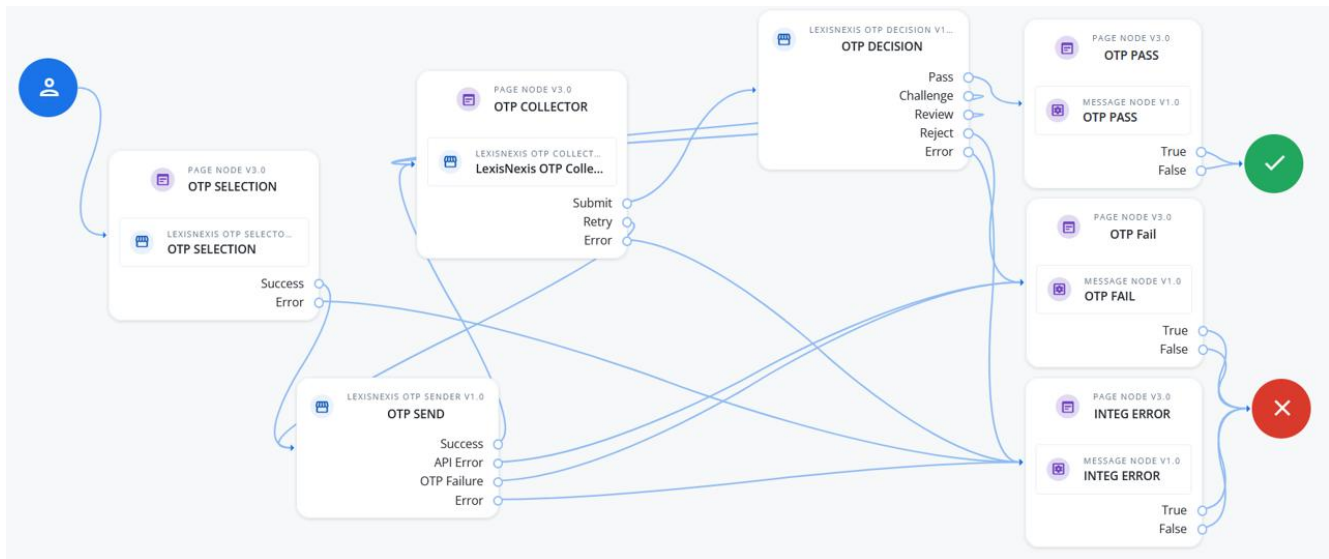
Cancel
OK

- Save the policy.
- Consult with LexisNexis professional services for a more comprehensive policy configuration.

AUTHENTICATION JOURNEY CONFIGURATION

Journey: LNRS-StepUp-OTP

This section provides the steps to configure a ForgeRock Journey with LexisNexis OTP nodes from the ForgeRock Identity Cloud Marketplace. LexisNexis OTP Nodes are used for Identity proofing as well as a multi-factor authentication (MFA) of a user. This example workflow displays the OTP Selector to the user, sends the OTP code, collects the OTP code value from the user and then validates if the value is correct.

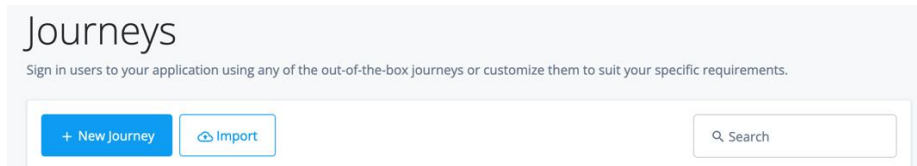


The flow is as follows:

- Page Node to display the OTP selection interface using the LexisNexis OTP Selector node. This configuration wraps the selector with page node so that detailed messages can be displayed to the user as part of the interface. Success from the selector will place the variable `otp.type_selected` into shared state.
- LexisNexis OTP Sender node will generate the OTP code to the end user. This node will inspect shared state for username and `otp.type_selected`. This combination of information defines the type of OTP to send to the defined user. The configuration of the node will define how additional attributes for the API request are fulfilled. For example, if the OTP type is email, then the Email Attribute will be fetched from either the User Directory or Shared state based on the configuration of the Attribute Source.
- Page Node to display the OTP Collection interface using the LexisNexis OTP Collector Node. This configuration wraps the selector with page node so that detailed messages can be displayed to the user as part of the interface.
- LexisNexis OTP Decision Node to determine if the OTP collected from the user is valid.
- Page Nodes with messages and a single OK button that will display the results and/or error conditions.

To configure the authentication journey, perform the following to configure the server:

1. From a workstation, launch a browser and navigate to the admin console.
2. Login with administrator credentials for a user that has permission to configure journeys
3. Upon login, select the realm to configure the journey. For the purposes of this instruction the default realm of **alpha** is assumed.
4. On the main dashboard display, select **Journeys** on the left-hand side of the display.
5. On the **Journeys** display, click the **+New Journey** button.



6. On the **New Journey** display, enter information as shown followed by the **Save** button.

New Journey ✕

Name
LNRS-OTP
Human-friendly name for this user flow.

Identity Object
Alpha realm - Users managed/alpha_user
Managed identity to authenticate using this journey.

Description (optional)
Step-Up Authentication integrated using LexisNexis OTP Nodes
Provide consumers of your flow a brief description of this flow.

Tags (optional)
Authentication ✕
Use tags to allow categorization and enhance searchability.

☐ Default journey for end users
Make this the default authentication journey for users in this realm.

☐ Override theme
Use a custom themed sign-in UI for this journey.

☐ Run journey for all users regardless of current session.

☐ Inner Journey
Only allow this journey to be initiated from within another journey.

Cancel Save

7. The result is the **Journeys > LNRS-StepUp-OTP** display. This is the interface to build up the journey. At this point, the journey will be built by drag-n-drop of Components on the left side of the screen. Each node in the policy will then be configured.

8. Build the OTP Method Selection Screen (e.g. Page Node), do the following:

- On the **Components Filter** on the left side of the interface, enter **page**. When the **Page Node** is displayed as a component, drag and drop the node into the authentication journey.
- When the **Page Node** properties are displayed on the right side of the interface, enter the following property values:

Node name	OTP SELECTION
Page Header	Two-Factor Authentication
Page Description	When ready, click the button below to receive a code to your device.

- On the **Components Filter** on the left side of the interface, enter **lexisnexis**. When the **LexisNexis OTP Selector** is displayed as a component, drag and drop it into the authentication journey into the **OTP SELECTION** page node.
- Select the **LexisNexis OTP Selector** Node component to display the configuration properties on the right side of the interface. Enter the following property values.

Send Email	Selected
Send SMS	Selected
Send Voice	Selected

9. Build the OTP Send, do the following:

- On the **Components Filter** on the left side of the interface, enter **lexisnexis**. When the **LexisNexis OTP Send** is displayed as a component, drag and drop it into the authentication journey.
- Select the **LexisNexis OTP Send** Node component to display the configuration properties on the right side of the interface. Enter the following property values.

Org ID	<ENTER ORG ID FROM DDP PORTAL>
API Key	<ENTER API KEY FROM DDP PORTAL>
Base URL	https://h-api.online-metrix.net/
Policy	OTP [NOTE: This is the configured policy name in DDP Portal]
OTP Length	6
OTP Expire	3
Email Title	Please, check out this email! It's your one time passcode.
Email Message	Your One Time Password is \${OTP}.\${LineBreak}We will never call you for this code.\${LineBreak}Your passcode will expire in \${ExpirePeriod} minutes.
SMS Message	Your One Time Password is \${OTP}.\${LineBreak}We will never call you for this code.\${LineBreak}Your passcode will expire in \${ExpirePeriod} minutes.
Attribute Source	User Directory
Email Attribute	mail
SMS Attribute	telephoneNumber
Voice Attribute	telephoneNumber

Note: In this example, the OTP Selector will fulfill the `otp.type_selected` shared state variable needed by the OTP Sender node. This variable defines the type of OTP that will be sent by the OTP Sender node.

Note: In this example, the authentication is assumed to be called via an inner tree evaluator. The journey that invokes this example authentication journey will insert the variable username into shared memory. The OTP Sender node requires this value.

Note: The attribute source is set to look in the user directory at runtime for the attribute to send to OTP. For example, if the OTP type is Email, then the OTP Sender node will look in the Identity Store, for the defined username, for the LDAP attribute mail.

10. Build the OTP Collector Screen (e.g. Page Node), do the following:

- On the **Components Filter** on the left side of the interface, enter **page**. When the **Page Node** is displayed as a component, drag and drop the node into the journey.
- When the **Page Node** properties are displayed on the right side of the interface, enter the following property values:

Node name	OTP COLLECTOR
Page Header	Identity Verification
Page Description	Please enter your one-time passcode.

- On the **Components Filter** on the left side of the interface, enter **lexisnexis**. When the **LexisNexis OTP Collector** is displayed as a component, drag and drop it into the journey into the **OTP COLLECTOR** page node.
- Select the **LexisNexis OTP Collector** Node component to display the configuration properties on the right side of the interface. Enter the following property values.

Message Body	Passcode was sent to \${otpDestination}. Please enter it below.
Help Text	One Time Passcode
OTP Error Message	Incorrect passcode entered, please try again.
OTP Blank Message	Blank passcode entered, please try again.

11. Build the OTP Decision, do the following:

- On the **Components Filter** on the left side of the interface, enter **lexisnexis**. When the **LexisNexis OTP Decision** is displayed as a component, drag and drop it into the journey.
- Select the **LexisNexis OTP Decision** Node component to display the configuration properties on the right side of the interface. Enter the following property values.

Org ID	<ENTER ORG ID FROM TMX PORTAL>
API Key	<ENTER API KEY FROM TMX PORTAL>

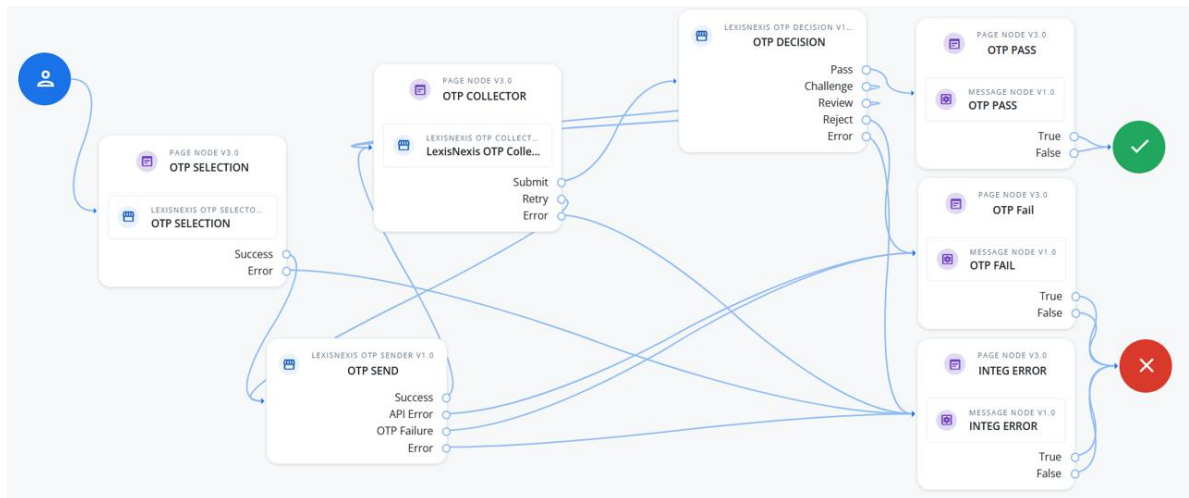
12. Build Message Nodes for OTP Pass, OTP Fail, and Integration Error to support testing. This can be accomplished with Message Nodes or Page Nodes with an OK Button Node. The nodes are meant to display outcomes to the user.

13. Link together the nodes of the journey

- To connect the nodes of the journey, click on the output dot of one node and then drag it to the input dot of another node. The following connections should be made.

Start	OTP SELECTION
OTP SELECTION (Success)	LexisNexis OTP Sender
OTP SELECTION (Error)	INTEGRATION ERROR
LexisNexis OTP Sender (Success)	OTP COLLECTOR
LexisNexis OTP Sender (API Error)	OTP FAIL
LexisNexis OTP Sender (OTP Failure)	OTP FAIL
LexisNexis OTP Sender (Error)	INTEGRATION ERROR
OTP COLLECTOR (Submit)	LexisNexis OTP Decision
OTP COLLECTOR (Retry)	OTP SELECTION
OTP COLLECTOR (Error)	INTEGRATION ERROR
LexisNexis OTP Decision (Pass)	OTP PASS
LexisNexis OTP Decision (Challenge)	OTP COLLECTOR
LexisNexis OTP Decision (Review)	OTP COLLECTOR
LexisNexis OTP Decision (Reject)	OTP FAIL
LexisNexis OTP Decision (Error)	INTEGRATION ERROR
OTP PASS (Next)	Success
OTP PASS (Error)	Failure
OTP FAIL (Next / Error)	Failure
INTEGRATION ERROR (Next / Error)	Failure

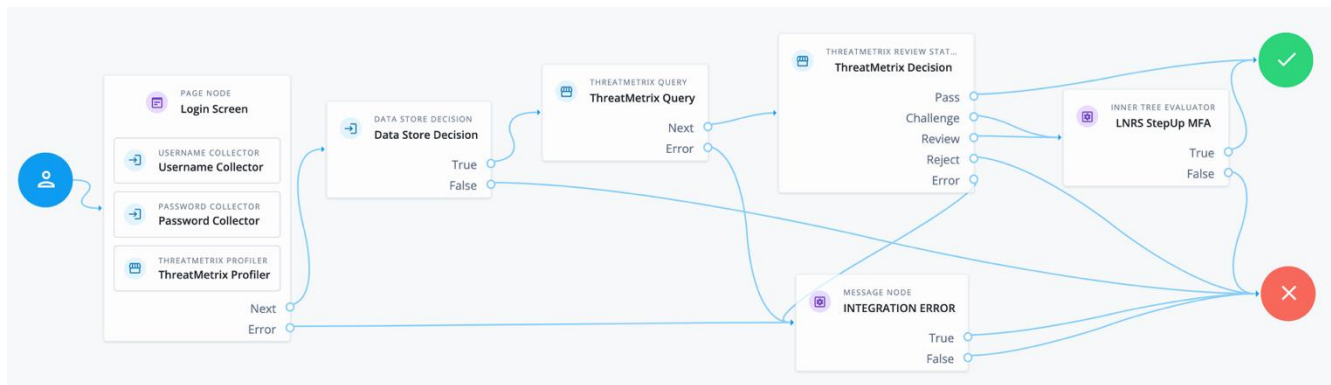
- At this point you should have the following



Journey: Integrate OTP into Existing Workflow

One-Time Passcode (OTP) technology is typically used to augment an orchestration for Identity Proofing or user MFA. The journey in the section **Journey: LNRS-StepUp-OTP** provides an example of the OTP workflow, which is an orchestrated workflow where the LexisNexis OTP Sender node has a dependency upon the attribute “username” being in shared state.

One typical use case for OTP is to integrate the nodes for step-up authentication following a risk assessment where the policy has rated the user as potential level of risk, thus requiring a second factor (e.g. step-up) authentication. Within ForgeRock journeys, this can be accomplished by inserting an Inner Tree Evaluator Node. In the example depicted below, there is a first factor authentication performed using the ForgeRock username/password collectors with a decision node, as well as LexisNexis ThreatMetrix Risk Assessment. The Inner Tree Evaluator Node is configured to handle risk assessment with an outcome of review or challenge.



For the purposes of brevity, the entire Login Journey will not be documented here, rather a simple documentation of the inner tree evaluator to integrate the LNRS-StepUp-OTP journey from the previous section.

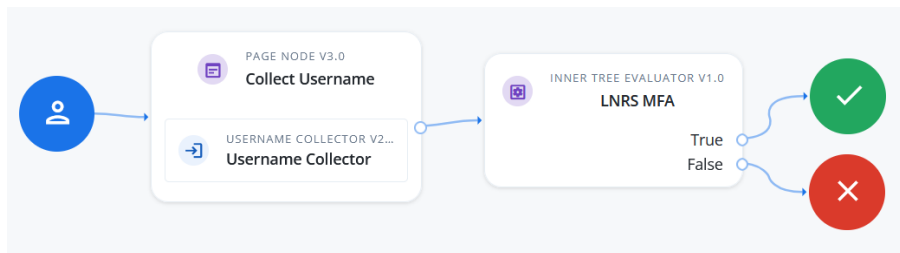
1. From a workstation, launch a browser and navigate to the ForgeRock Identity Cloud.
2. Login with administrator credentials for a user that has permission to configure journeys
3. Upon login, select the realm to configure the journey. For the purposes of this instruction the default realm of **alpha** is assumed.
4. On the main dashboard display, select **Journeys** on the left-hand side of the display.
5. On the **Journeys** display, select the existing journey to modify.
6. To leverage the OTP journey, add an Inner Tree Evaluator node.
 - On the **Components Filter** on the left side of the interface, enter **inner**. When the **Inner Tree Evaluator** node is displayed as a component, drag and drop an instance into the journey. This node will be used to display call the OTP Step Up journey. Enter the following property values.

Node name	LNRS StepUp MFA
Tree Name	LNRS-StepUp-OTP
 - Connect the nodes of the journey and save.

Journey: LNRS OTP Authentication

One-Time Passcode (OTP) technology is typically used to augment an orchestration for Identity Proofing or user MFA. The ForgeRock Journey in the section **Journey: LNRS-StepUp-OTP** provides an example of the OTP workflow, which is an orchestrated workflow where the LexisNexis OTP Sender node has a dependency upon the attribute “username” being in shared state.

A simple framework to exercise the Journey is to configure a page node with a username collector which will fulfill the dependency to have the attribute “username” being in shared state. Then an Inner Tree Evaluator Node can integrate the OTP journey for authentication final outcome.



Perform the following.

1. From a workstation, launch a browser and navigate to the ForgeRock Identity Cloud.
2. Login with administrator credentials for a user that has permission to configure journeys
3. Upon login, select the realm to configure the journey. For the purposes of this instruction the default realm of **alpha** is assumed.
4. On the main dashboard display, select **Journeys** on the left-hand side of the display.
5. On the **Journeys** display, click the **+New Journey** button.
6. On the **New Journey** display, enter information for the journey followed by the **Save** button.

Name	LNRS-OTP-Authentication
Identity Object	Select Alpha Users
Description	Enter a short description
Tags	Authentication

7. The result is the **Journeys > LNRS-OTP-Authentication** display. This is the interface to build up the journey. At this point, the journey will be built by drag-n-drop of Components on the left side of the screen. Each node in the policy will then be configured.
8. Build the Collector Username Screen (e.g. Page Node), do the following:
 - On the **Components Filter** on the left side of the interface, enter **page**. When the **Page Node** is displayed as a component, drag and drop the node into the authentication tree.
 - When the **Page Node** properties are displayed on the right side of the interface, enter the following property values:

Node name	Collect Username
Page Header	OTP Authentication
 - On the **Components Filter** on the left side of the interface, enter **username**. When the **Username Collector** is displayed as a component, drag and drop the **Username Collector** into the authentication tree into the **Login Screen** page node.

9. To leverage the OTP authentication tree, add an Inner Tree Evaluator node.

- On the **Components Filter** on the left side of the interface, enter **inner**. When the **Inner Tree Evaluator** node is displayed as a component, drag and drop an instance into the authentication tree. This node will be used to display call the OTP Step Up authentication tree. Enter the following property values.

Node name	Inner Tree - MFA
Tree Name	LNRS-StepUp-OTP

- Connect the nodes of the authentication tree and save the authentication tree.
- The result should look like the following:

